

Expectation Decider

Insights Report

Dataset: 450 Students | 7 Probability Topics

Q1 — Basic Probability Events

Three probability events were computed from the dataset of 450 students:

Event	Favorable	Total	Formula	P(E)
Student Passes Exam	312	450	312 / 450	0.6933
Participates in Group Disc.	211	450	211 / 450	0.4689
Attendance > 75%	226	450	226 / 450	0.5022

Insight: 69.33% of students passed — significantly above 50%. Only 46.89% join group discussions, yet attendance above 75% stands at 50.22%, suggesting many attend regularly but don't participate actively in discussions.

Q2 — Empirical vs. Theoretical Probability

Empirical Probability (from actual data):

$$P(\text{Pass}) = \frac{\text{Pass Students}}{\text{Total Students}}$$
$$= \frac{312}{450} = 0.6933$$

Theoretical Probability (equal-likelihood assumption):

$$S = \{\text{Pass}, \text{Fail}\} \Rightarrow \text{Favorable} = 1, \text{ Total} = 2$$
$$P(\text{Pass}) = \frac{1}{2} = 0.5000$$

Type	P(Pass)	Basis
Empirical	0.6933 (69.33%)	Real dataset — 312 out of 450 passed
Theoretical	0.5000 (50.00%)	Equal chance assumption (Pass or Fail)
Difference	+0.1933	Real-world pass rate is 19.33% higher

Insight: The empirical pass rate (69.33%) is nearly 20 percentage points above the theoretical 50%. Real-world factors — study effort, attendance, prior knowledge — push outcomes well above a pure coin-flip scenario.

Q3 — Random Variable & Binomial Distribution

X = Number of students passing out of $n = 3$ randomly selected. $p = 312/450 = 0.6933$, $q = 1 - p = 0.3067$

Binomial PMF:

$$P(X = k) = C(n, k) \times p^k \times (1-p)^{(n-k)}$$

X	C(3,X)	p^X	$(1-p)^{(3-X)}$	P(X)	Meaning
0	1	$0.6933^0 = 1$	$0.3067^3 = 0.0288$	0.0288	All 3 fail
1	3	$0.6933^1 = 0.6933$	$0.3067^2 = 0.0941$	0.1956	Exactly 1 passes
2	3	$0.6933^2 = 0.4807$	$0.3067^1 = 0.3067$	0.4423	Exactly 2 pass
3	1	$0.6933^3 = 0.3333$	$0.3067^0 = 1$	0.3333	All 3 pass

$$\text{Mean} = n \times p = 3 \times 0.6933 = 2.08$$

$$\text{Variance} = n \times p \times q = 3 \times 0.6933 \times 0.3067 = 0.6379$$

Insight: In any random group of 3 students, on average 2 will pass. The most likely outcome is 'exactly 2 pass' (44.23%). There's only a 2.88% chance all 3 fail, confirming the strong pass trend in this batch.

Q4 — Venn Diagram: Study Hours & Attendance

Set A = Students with study_hours > 10 | Set B = Students with attendance > 80%

Region	Count	Percentage	Description
Only A (Study > 10 only)	200	44.44%	High study, attendance <= 80%
A n B (Both conditions met)	140	31.11%	High study AND high attendance
Only B (Attendance > 80 only)	40	8.89%	High attendance, study <= 10 hrs
Neither A nor B	70	15.56%	Low study AND low attendance

Calculated Probabilities:

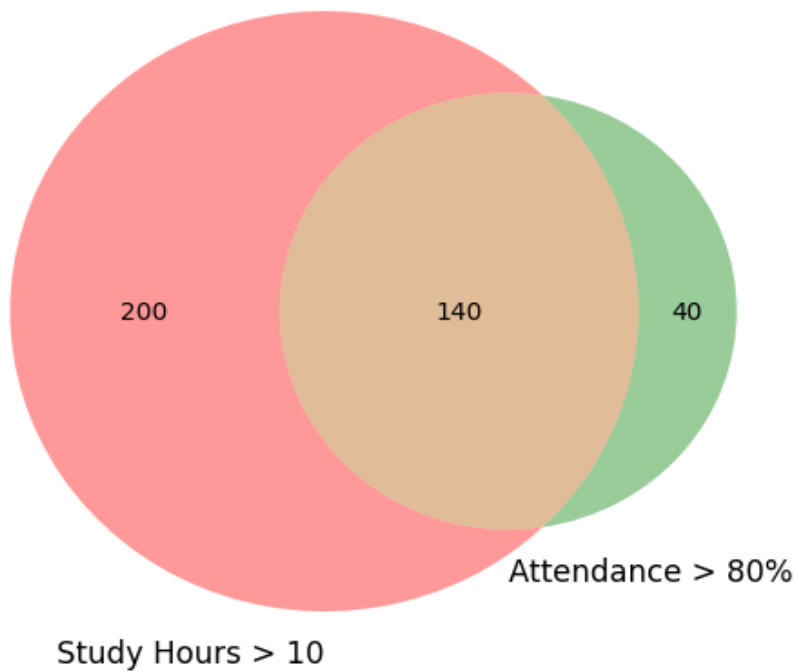
$$P(A) = (200+140) / 450 = 340/450 = 0.7556$$

$$P(B) = (140+40) / 450 = 180/450 = 0.4000$$

$$P(A \cap B) = 140 / 450 = 0.3111$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.7556 + 0.4000 - 0.3111 = 0.8444$$

Venn Diagram of Student Groups



Insight: 31.11% of students both study >10 hrs/week AND attend >80% of classes — this group likely has the highest pass rate. The 15.56% who do neither form the highest-risk group for failure.

Q5 — Contingency Table & Probability Calculations

Group Discussion	Fail	Pass	Row Total
No	84	155	239
Yes	54	157	211
Column Total	138	312	450

Joint Probability — $P(\text{Group Discussion} = \text{Yes AND Pass})$

$$= \text{Favorable (Yes \& Pass)} / \text{Total} = 157 / 450 = 0.3489$$

Marginal Probability — $P(\text{Pass})$

$$= \text{Total Pass} / \text{Total Students} = 312 / 450 = 0.6933$$

Conditional Probability — $P(\text{Pass} | \text{Group Discussion} = \text{Yes})$

$$\begin{aligned} &= P(\text{Yes AND Pass}) / P(\text{Yes}) = (157/450) / (211/450) \\ &= 157 / 211 = 0.7441 \end{aligned}$$

Insight: Students who participate in group discussions pass at 74.41% vs 64.85% among non-participants (155/239). Participating in group discussions is associated with a ~10 percentage point higher pass rate.

Q6 — Independence & Mutual Exclusivity

Test for Independence: $P(\text{Pass} \mid \text{Group Disc.}) = P(\text{Pass})$?

$$P(\text{Pass} \mid \text{Yes}) = 0.7441$$

$$P(\text{Pass}) = 0.6933$$

$$0.7441 \neq 0.6933 \Rightarrow \text{Events are DEPENDENT}$$

Test for Mutual Exclusivity: Can a student both participate AND pass?

$$\text{Yes} - 157 \text{ students did both} \Rightarrow \text{Events are NOT Mutually Exclusive}$$

Insight: Since the probability of passing changes when we know a student participated in group discussions (0.7441 vs 0.6933), the two events are dependent. Group discussion participation is a positive influencing factor on exam outcomes.

Q7 — Bayes' Theorem: $P(\text{Pass} \mid \text{High Attendance})$

Given Information:

Symbol	Value	Meaning
$P(H \mid \text{Pass})$	0.70	70% of students who passed had attendance > 80%
$P(H \mid \text{Fail})$	0.40	40% of students who failed had attendance > 80%
$P(H)$	0.60	60% of all students had high attendance
$P(\text{Pass})$	$312/450 = 0.6933$	From the dataset
$P(\text{Fail})$	$138/450 = 0.3067$	From the dataset

Bayes' Theorem Formula:

$$P(\text{Pass} \mid H) = [P(H \mid \text{Pass}) \times P(\text{Pass})] / P(H)$$

Substitution:

$$\begin{aligned} P(\text{Pass} \mid H) &= [0.70 \times 0.6933] / 0.60 \\ &= 0.4853 / 0.60 \\ &= 0.8089 = 80.89\% \end{aligned}$$

Probability	Value	Interpretation
$P(\text{Pass})$ — prior	0.6933	Base pass rate without any condition
$P(\text{Pass} \mid \text{High Att.})$ — posterior	0.8089	Pass rate knowing attendance > 80%

Probability	Value	Interpretation
Uplift	+11.56 pp	High attendance boosts pass probability

Insight: A student with high attendance (>80%) has an 80.89% chance of passing — 11.56 percentage points above the overall pass rate. High attendance is the strongest single predictor of passing in this analysis.

Final Summary — What Most Affects the Probability of Passing?

Across all seven probability topics applied to the 450-student dataset, three key factors clearly drive the probability of passing the final exam:

Rank	Factor	P(Pass) With Factor	P(Pass) Without	Impact
1	High Attendance (>80%)	80.89% — Bayes	~58-60% (implied)	Strongest: +11.56 pp
2	Group Discussion Participation	74.41% — Conditional	64.85% — No-discussion	Moderate: +9.56 pp
3	High Study Hours (>10 hrs/wk)	Venn overlap group	Neither-group high risk	Structural factor

Conclusion

The most influential factor is high attendance (>80%), which raises the pass probability to 80.89% via Bayes' Theorem. Group discussion participation adds another ~10 pp advantage through conditional probability. Students who combine both habits — attend regularly AND participate actively — form the highest-performing segment (31.11% of the batch: 140 students satisfy both study > 10 hrs and attendance > 80%). The overall batch performs well above random chance (69.33% vs theoretical 50%), confirming that consistent effort and engagement are the true 'Expectation Deciders.'